



Reg. No.

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**BCACACS 101**

**First Semester B.C.A. Degree Examination, December 2025/January 2026  
(SEP) (2024-25 Batch Onwards)  
FUNDAMENTALS OF COMPUTERS**

Time : 3 Hours

Max. Marks : 80

**Note :** Answer *any ten* questions from Part – A and *one full* question from *each* unit of Part – B.

**PART – A**

1. a) Differentiate RAM and ROM. (10×2=20)  
b) What is input device ? Give two examples.  
c) Differentiate Program counter and Instruction Register.  
d) Define System Software. Give an example.  
e) List any 2 symbols used in flow charts along with their purpose.  
f) Differentiate compilers and interpreters.  
g) Perform  $(100101.11)_2 + (11011.101)_2$ .  
h) Write the BCD and Excess-3 Codes for 257.  
i) Write 1's and 2's Complement of  $(11001100)_2$ .  
j) Write truth table and logic expression of NOR gate.  
k) Draw the logic circuit for  $F(X, Y, Z) = X'Y + X'Z$ .  
l) Write the general structure of 3 variables K-Map.

**PART – B**

**Unit – I**

2. a) Write a note on :  
i) Mainframes  
ii) Supercomputers.  
b) Write the characteristics of First-Generation Computers.  
c) List and explain characteristics of Computers.

**(4+5+6)**

**P.T.O.**



3. a) Differentiate LCD and CRT monitors.  
b) Explain the applications of Computers in different fields.  
c) Draw the block diagram of a Computer system and explain the components of a Computer system. (4+5+6)

### Unit – II

4. a) Define Operating System. List and explain basic functions of an Operating System.  
b) List and explain the advantages of high level languages.  
c) Briefly explain different phases of the Program Development Cycle. (5+5+5)
5. a) Write a note on :  
i) Word Processors  
ii) Presentation Software.  
b) What is System Utility ? Explain any four functions of System Utility.  
c) Explain Features of Good Programming Language. (4+5+6)

### Unit – III

6. a) State the Postulates of Boolean algebra.  
b) Perform following Subtraction using 1's and 2's complement methods  
·  $(10011)_2 - (1001)_2$ .  
c) Perform following conversion :  
i)  $(1101.11)_2 = ( )_{10}$   
ii)  $(37)_8 = ( )_{16}$   
iii)  $(45)_{10} = ( )_2$  (4+5+6)



7. a) State and Prove De-Morgan's theorems for two variables.  
b) Perform following Subtraction using 9's and 10's complement methods.  
 $(8052)_{10} - (3250)_{10}$   
c) Perform following conversion.  
i)  $(111101.10)_2 = ( )_8$   
ii)  $(45.B3)_{16} = ( )_2$   
iii)  $(789)_{10} = ( )_2$  (4+5+6)

#### Unit – IV

8. a) Express the Boolean function  $F = XY' + X'Z$  as sum of minterm and product of maxterm.  
b) How do you get complement of a function ? Find the complement of :  
i)  $F1 = x'y'z' + x'y'z$   
ii)  $F2 = x(y'z' + yz)$   
c) Write a note on K-Map. Using K-Map simplify the following expression  
 $F(w, x, y, z) = \Sigma(0, 1, 2, 4, 5, 6, 8, 9, 12, 13, 14)$ . (5+5+5)
9. a) Prove that NAND is universal gate.  
b) Using Boolean Theorems and postulates, prove the following :  
i)  $x + x'y = x + y$ .  
ii)  $x'y'z + x'yz + xy' = x'z + xy'$ .  
c) Implement Boolean function  $F = x'y'z + x'yz + yz'$  with basic gate and also write the truth table. (5+5+5)
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